# Galaba Vamsi

**Software Engineer (Full-stack, AI Agents), Intern / New Grad (graduating May 2027)**
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## SUMMARY

Mechatronics at IIT Bhilai. I ship full-stack products and AI agents: co-founder of Human Slop (live on the web and Google Play) and Robotropic; Charter, an MCP commerce agent; Pehchaan, top 10 of 4,500+ teams at RAKSHAM. 15 merged open-source PRs (11 code) and a maintained PyPI package.

## EDUCATION

**B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai** Aug 2023 - May 2027
**Coursework:** DSA, OOP, AI and ML, Deep Learning for Computer Vision, Digital Image Processing.

## EXPERIENCE

**Co-founder, Robotropic** Physical-AI startup, with Varun Rao | robotropic.com Oct 2026 - Present
- Building Harness, a workspace where a factory's team and AI agents change robot skills safely: each change is checked against safety limits, rolled out cell by cell, and recorded. I lead the simulated factory floor and the change gateway.
- Built robotropic.com: three.js lidar factory scene, early-access list on Cloudflare Workers and D1.

**Co-founder, Human Slop** Anti-AI social app, with Varun Rao | humanslop.in 2026 - Present
- Expo (React Native) app for web and Android, Fastify API, PostgreSQL on Supabase, deployed on Render and Cloudflare; live on the web and Google Play (50+ downloads).
- Built a signup biometric check and an editor keystroke-timing score (heuristics, not proof of authorship).

**Research Intern, IIT Bhilai** Advisor: Dr. Arzad Alam Kherani May 2025 - Jul 2025
- Built RIS-SIM, an open-source Python radio-link emulator with USRP-style SDR interfaces (later rebuilt as sremu); co-author, IEEE INDICON 2025.

## PROJECTS

**Pehchaan** (team of 2; Top 10, RAKSHAM (IIT Delhi, I4C and Amazon), 2026) Voice-clone payment-fraud defence for UPI; I led ML and backend. github.com/Galabavamsi/pehchaan
- FastAPI risk API that fuses signals by log-odds under 7 hard guardrails (p95 decision latency 3.8 ms, local benchmark); React payer, relative and analyst demo over WebSockets; Android SDK demo on two phones; 120 automated tests.
- Consent registry that refuses to dial, record or clone without saved consent; fine-tuned detector at 4.8% EER vs 34.3% off the shelf.

**Charter** (Razorpay AI Buildathon, 2026) Agentic commerce: chat, voice, and a 10-tool MCP server call one first-party API, so agents get no side door; only versioned policy changes money (frozen INR quotes, Razorpay test mode, PostgreSQL row-level security). TypeScript, Fastify, React; deployed to AWS Lightsail via GitHub Actions with OIDC. charter.humanslop.in (live, test mode) | github.com/Galabavamsi/charter

**sremu** (2026, MIT, built with Claude Code) Python IQ-level radio emulator with a ZeroMQ data plane that unmodified GNU Radio 3.10 talks to; timestamped commands act on an exact, logged sample; about 2,140 tests; bit-identical on Windows and Linux; benchmarked on AWS; sole-author preprint draft, arXiv submission pending. github.com/Galabavamsi/sremu

## OPEN-SOURCE CONTRIBUTIONS

**HFlow, Hebbian Robotics (YC S26)** github.com/Hebbian-Robotics/hflow Sep 2026 - Oct 2026
- 8 merged PRs (Python, Apache Arrow) to an SDK that verifies robot training data: LeRobot and robot-log conversion now keeps data whole or refuses it with a reason. Blocked silent 12- to 10-bit depth loss (PR #400), typed Arrow columns from every message (PR #657), fixed video-cache collisions (PR #353).

**Inspect Robots, Robocurve (YC S26)** github.com/robocurve/inspect-robots Sep 2026 - Oct 2026
- Evaluation framework for embodied-AI policies. 2 merged PRs: a configurable LLM-API retry policy across 5 wire protocols (PR #476) and unknown git state no longer logged as clean (PR #475); 3 open, including checkpoint/resume for long eval sets (PR #514).
- Wrote and maintain inspect-robots-wandb, a Weights & Biases plugin (pypi.org/project/inspect-robots-wandb): 100% coverage gate, CI on Python 3.10-3.13, trusted publishing.

**Cerulion, Rust robot runtime** 1 merged CLI fix (PR #216), 4 documentation fixes | github.com/cerulion-inc/cerulion Oct 2026

## SKILLS

**Languages:** TypeScript/JavaScript, Python, SQL, Rust
**Backend and frontend:** Node.js (Fastify), FastAPI, React, React Native (Expo), PostgreSQL (row-level security), Supabase, WebSockets, REST APIs
**AI:** LLM APIs, AI agents and tool use, MCP (Model Context Protocol), agent evaluation, PyTorch
**Infrastructure:** Docker, GitHub Actions (OIDC), AWS Lightsail, Cloudflare Workers and D1, Render, Linux, Git, pytest
**AI-assisted development:** Claude Code; I set the design and verify results with tests and independent references.

## ACHIEVEMENTS

- **Top 10 (Grand Jury round) of 4,500+ teams nationwide**, RAKSHAM AI Cybersecurity Hackathon (IIT Delhi, I4C and Amazon), Oct 2026; Pehchaan, team of 2.
- **5th of 232 teams**, WEAR Dataset Challenge @HASCA 2026 (Kaggle); **All India Rank 200** (2026) and **700** (2025), Amazon ML Challenge (team); **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team); **1st place**, IIT Bhilai Web Development Competition.
- Prime Minister's Scholarship Scheme (PMSS) scholar, 2024; Subroto Mukherjee Scholarship (IAFBA), 2023; Top 1% in JEE Advanced; attended YC Startup School, Bengaluru.

## PUBLICATIONS

- **Co-author (8th of 15)**, *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306; SPAIS Workshop at CoRL 2026 (poster).
- **Co-author**, 2 RIS papers: IEEE INDICON 2025 and *Ad Hoc Networks* (Elsevier), 2026.